

Antonio De Leon

Santa Cruz, CA | jaguir26@ucsc.edu | antonio-de-leon.com | github.com/AntonioAPDL | linkedin.com/in/antonioapdl

Professional Summary

Statistics Ph.D. candidate and applied quantitative researcher with experience in probabilistic forecasting, statistical learning, Bayesian computation, and reproducible model evaluation. Builds analytical workflows in Python, R, SQL, and C++ across financial, sensor, image, and environmental data, from data construction through diagnostics, software, and communication.

Education

Ph.D. in Statistics, University of California, Santa Cruz Expected December 2026

GPA: 3.82/4.00; candidacy completed. Research: probabilistic forecasting, model evaluation, and statistical software.

M.Sc. in Economic Theory, ITAM 2018–2020

GPA: 96.7/100.

B.Sc. in Applied Mathematics, ITAM 2014–2018

GPA: 97.1/100.

Relevant coursework and training: probability, statistical inference, regression, statistical learning, Bayesian inference, time series, stochastic processes, decision theory, causal inference, and impact evaluation.

Technical and Quantitative Skills

Programming: Python, R, SQL, C++, Bash, Git, LaTeX.

Statistical and machine-learning methods: probability, regression, Bayesian inference, time-series and state-space models, statistical learning, Monte Carlo methods, variational inference, experimental design, causal inference and impact-evaluation training, model diagnostics, calibration, and proper scoring.

Systems and tools: Linux, AWS (EC2, EMR, S3), Stan, Rcpp, CRAN, GitHub Actions, Tableau, DBeaver.

Languages: Spanish, English, German.

Professional Experience

Delos Financial Technologies, San Rafael, CA (Remote) May 2025–Sep. 2025

Quantitative Researcher (Contract); Python, Git, Bash, C++, AWS, Stan

- Built Python model-evaluation workflows for residual diagnostics, goodness-of-fit analysis, cohort summaries, and backtest review across financial-modeling experiments.
- Implemented baseline and extended forecasting specifications, automated AWS experiment runs, and standardized versioned outputs for reproducible model comparison.

UCSC Statistics Department, Santa Cruz, CA Feb. 2024–Present

Computer Systems Coordinator; Linux, Bash, Python/R, Networking, User Support

- Administer Linux research infrastructure for faculty and graduate students, including software stacks, storage, upgrades, access, and incident response.
- Write automation scripts and documentation for onboarding, resource usage, and reproducible computing; coordinate with central IT on research infrastructure planning.

NeatLeaf Inc., Scotts Valley, CA May 2021–Sep. 2022

Data Analyst; Python, SQL, Git, Bash, Linux, C++, DBeaver

- Built Python and SQL pipelines for greenhouse robot and sensor telemetry used in operational monitoring and downstream analysis.
- Analyzed spatiotemporal microclimate anomalies and robot/sensor calibration logs to improve data-quality review and system reliability.

Banco de México (BANXICO), Mexico City, Mexico Aug. 2018–Dec. 2019

Data Analyst; Python, R, SQL, Git

- Built Python/R/SQL pipelines for banknote image and operational datasets used in monitoring, anomaly analysis, and model development.
- Developed anomaly-classification diagnostics and financial-volume forecasting prototypes for operational planning and review.

Selected Research and Software

exdqIm – CRAN R package v1.1.0 2026

Software article submitted to the Journal of Statistical Software

- Co-developed a public package for Bayesian dynamic and static quantile models, including MCMC and variational inference, forecasting, diagnostics, tests, documentation, and reproducible examples.
- Added evaluation utilities and selected C++ paths, with R reference implementations for validation and portability.

Bayesian Quantile Deep Echo State Networks

2024–Present

Working paper on nonlinear time-series forecasting

- Developed probabilistic time-series models using fixed nonlinear recurrent features, regularization, MCMC, and variational inference.
- Designed simulation and rolling held-out studies, including a selected-panel electricity-price forecasting comparison with 42 region/fold evaluations.

Bayesian Quantile-Based Correction and Synthesis of Hydrologic Products

2021–Present

arXiv preprint; manuscript submitted to Environmetrics

- Integrated USGS observations, retrospective products, ensemble forecasts, and covariates from multiple agencies into origin-specific, version-consistent datasets.
- Evaluated probabilistic forecasts across five origins and seven fitted quantile levels using CRPS and quantile loss, keeping post-origin observations reserved for verification.

Selected Publications

De Leon, A., Barata, R., Prado, R., & Sansó, B.

exdqlm: An R Package for Estimation and Analysis of Flexible Dynamic Quantile Linear Models.

Submitted to the *Journal of Statistical Software*; CRAN R package v1.1.0; arXiv:2607.22760.

De Leon, A., Prado, R., & Sansó, B.

Bayesian Quantile Deep Echo State Networks.

Working paper on probabilistic time-series modeling with fixed reservoir features, MCMC, variational inference, and held-out forecast evaluation.

De Leon, A., Prado, R., & Sansó, B.

Mean-Tilted Intervals: A Generalized-Bayes Approach to Fixed-Content and Tolerance Intervals.

arXiv preprint on interval estimands, generalized-Bayes computation, and tolerance-interval actions; arXiv:2607.26098.

De Leon, A., Prado, R., & Sansó, B.

Bayesian Quantile-Based Correction and Synthesis of Hydrologic Products.

arXiv preprint; manuscript submitted to *Environmetrics*; arXiv:2608.11222.

De Leon, A., & Lobato, I. N.

Evidence of non-fundamentality in OECD capital stocks.

Empirical Economics 67, 761–772 (2024).

Awards, Teaching, and Service

Awards: Best Poster Prize, ISBA 2026 World Meeting; EnviBayes Student Paper Competition Winner, 2026; ISBA 2026 Junior Researcher Travel Award; UC Regents Fellowship, UCSC Statistics, 2021; Baillères Fellowship, ITAM, 2014–2020.

Teaching and mentoring: Graduate Student Instructor for *Data Visualization*; Teaching Assistant for *Probability Theory, Classical and Bayesian Inference, Statistics*, and related courses; ASA DataFest mentor for student teams in modeling, visualization, and presentation.

Service and training: TA Wiki maintainer and workshop contributor for Git, GitHub, collaborative documentation, and departmental teaching resources; contributed to an anonymous, IRB-exempt TA Wiki survey and follow-up resource improvements.